Complete Proof of Proposition 1

Companion note to K.-H. Lee, H.-H. Choi, and J.-R. Lee, "Semantic Multiple Access via Structured Shared-Private Embeddings in Mobile Networks," IEEE Transactions on Mobile Computing (under review, 2026)

The paper states Proposition 1 and gives the structure of its proof. This note gives the complete derivation of the three limits and of the optimality property. The notation is that of Section IV of the paper, receiver names and acronyms are those of the paper, and equation numbers in parentheses refer to the paper.

Proposition 1 *The affinity-aware LMMSE receiver is* $\hat{\mathbf{E}} = (\mathbf{F}^\star \otimes \mathbf{I}_d)\tilde{\mathbf{E}}$ *with*

$$\mathbf{F}^\star \;=\; \mathbf{C}_x\boldsymbol{\Gamma}^\top\big(\boldsymbol{\Gamma}\mathbf{C}_x\boldsymbol{\Gamma}^\top + \mathbf{C}_n\big)^{-1}, \tag{10}$$

and per-user mean-square error $d\,[\mathbf{C}_x - \mathbf{F}^\star\boldsymbol{\Gamma}\mathbf{C}_x]_{uu}$ *from the Wiener error covariance. It has the following properties.*

(i) *As* $\sigma^2 \to 0$, $\mathbf{F}^\star \to \boldsymbol{\Gamma}^{-1}$, *which is the SimR receiver (8).*

(ii) *As* $\beta_{uv} \to 0$ *for all pairs,* $\mathbf{F}^\star \to \operatorname{diag}\big(h_u^2/(h_u^2 + d\sigma^2)\big)$, *which is independent per-user Wiener filtering.*

(iii) *As* $\beta_{uv} \to 1$ *for all pairs,* $\mathbf{B}$ *collapses to rank one and* $\mathbf{F}^\star$ *averages the* U *front-end outputs with maximal-ratio weights* h_v^2. *This is SimC up to a scalar Wiener gain.*

(iv) $\mathbf{F}^\star$ *minimizes the mean-square error among all linear receivers of the form* $\mathbf{F} \otimes \mathbf{I}_d$ *and hence dominates both SimR and SimC at every* $(\mathbf{B}, \rho)$.

Proof

The Wiener filter within the class of receivers of the form $\mathbf{F} \otimes \mathbf{I}_d$ is

$$\mathbf{F}^\star = \mathbb{E}[\check{\mathbf{x}}\check{\mathbf{y}}^\top]\,\mathbb{E}[\check{\mathbf{y}}\check{\mathbf{y}}^\top]^{-1},$$

and substituting the joint statistics of the front-end model gives the closed form quoted in the proposition.

Cancellation at vanishing noise

Letting $\mathbf{C}_n \to \mathbf{0}$ gives

$$\mathbf{F}^\star \to \mathbf{C}_x\boldsymbol{\Gamma}^\top(\boldsymbol{\Gamma}\mathbf{C}_x\boldsymbol{\Gamma}^\top)^{-1} = \mathbf{C}_x\boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}^{-\top}\mathbf{C}_x^{-1}\boldsymbol{\Gamma}^{-1} = \boldsymbol{\Gamma}^{-1},$$

where every factor is invertible, because $\mathbf{B} = \operatorname{diag}(1 - a_u^2) + \mathbf{a}\mathbf{a}^\top$ is the sum of a positive definite diagonal matrix and a positive semidefinite rank-one matrix and is therefore positive definite whenever every $a_u < 1$. The receiver therefore approaches the SimR receiver, which inverts the coupling exactly.

Per-user Wiener filtering at zero affinity

Substituting $\mathbf{B} = \mathbf{I}$ gives $\boldsymbol{\Gamma} = \mathbf{I}$ and $\mathbf{C}_n = \sigma^2\mathbf{H}^{-2}$, so

$$\mathbf{F}^\star = \frac{1}{d}\left(\frac{1}{d}\mathbf{I}_U + \sigma^2\mathbf{H}^{-2}\right)^{-1} = \operatorname{diag}\left(\frac{h_u^2}{h_u^2 + d\sigma^2}\right).$$

This is the scalar Wiener filter of a single-user channel whose per-dimension SNR is $h_u^2/(d\sigma^2)$, which is the expected result once the users no longer interfere.

Coherent combining at unit affinity

As $\mathbf{B} \to \mathbf{1}\mathbf{1}^\top$ both $\mathbf{\Gamma}\mathbf{C}_x\mathbf{\Gamma}^\top$ and $\mathbf{C}_n$ collapse onto the rank-one span of $\mathbf{H}^{-1}\mathbf{1}$. On that span the Wiener solution is the scalar problem of estimating one common signal from its maximal-ratio combination, whose weights are h_v^2. By the matrix inversion lemma, (10) equals $\mathbf{F}^\star = (d\sigma^2\mathbf{B}^{-1} + \mathbf{HBH})^{-1}\mathbf{H}^2$. The share coefficients may differ across users. Let $\xi = \max_u(1-a_u^2)$, so that $\beta_{uv} = a_u a_v \to 1$ for all pairs exactly when $\xi \to 0$. Equal affinities, $a_u^2 = \beta$ for every u, are the special case $\xi = 1-\beta$. The matrix $\mathbf{B} = \mathrm{diag}(1-a_u^2)+\mathbf{a}\mathbf{a}^\top$ is a rank-one update of a diagonal matrix. By eigenvalue interlacing [1], its $U-1$ smallest eigenvalues lie in $(0, \xi]$. Its largest eigenvalue μ_1 tends to U, and its unit eigenvector $\mathbf{w}$, signed so that $\mathbf{1}^\top\mathbf{w} > 0$, tends to $\mathbf{1}/\sqrt{U}$, because $\mathbf{B} \to \mathbf{1}\mathbf{1}^\top$. Split $d\sigma^2\mathbf{B}^{-1} + \mathbf{HBH} = \mathbf{K}_\perp + \mathbf{K}_\parallel$ with $\mathbf{K}_\perp = d\sigma^2(\mathbf{B}^{-1} - \mathbf{w}\mathbf{w}^\top/\mu_1)$ and $\mathbf{K}_\parallel = d\sigma^2\mathbf{w}\mathbf{w}^\top/\mu_1 + \mathbf{HBH}$. The matrix $\mathbf{K}_\perp$ acts on the complement of $\mathbf{w}$ with eigenvalues of at least $d\sigma^2/\xi$, while $\mathbf{K}_\parallel$ stays bounded and $\mathbf{w}^\top\mathbf{K}_\parallel\mathbf{w} \geq d\sigma^2/\mu_1 > 0$. The block inverse with respect to $\mathbf{w}$ and its complement then gives

$$(\mathbf{K}_\perp + \mathbf{K}_\parallel)^{-1} = \frac{\mathbf{w}\mathbf{w}^\top}{\mathbf{w}^\top\mathbf{K}_\parallel\mathbf{w}} + O(\xi).$$

With $\mathbf{h} = \mathbf{H}\mathbf{1}$, the denominator tends to $\big(d\sigma^2 + (\mathbf{1}^\top\mathbf{h})^2\big)/U$, so $\mathbf{F}^\star \to \mathbf{1}\mathbf{1}^\top\mathbf{H}^2/\big((\mathbf{1}^\top\mathbf{h})^2 + d\sigma^2\big)$. Every row is the maximal-ratio combiner with weights h_v^2 followed by the scalar Wiener gain, which is the SimC behavior.

Optimality within the receiver class

Property (iv) is the orthogonality principle $\mathbb{E}[(\check{\mathbf{x}} - \mathbf{F}^\star\check{\mathbf{y}})\check{\mathbf{y}}^\top] = \mathbf{0}$, which characterizes the minimizer of the mean-square error within the class of receivers of the form $\mathbf{F} \otimes \mathbf{I}_d$. The SimR and SimC receivers are both of this form, with $\mathbf{F} = \mathbf{\Gamma}^{-1}$ and with $\mathbf{F}$ equal to a rank-one matrix of maximal-ratio rows respectively, so both are dominated by $\mathbf{F}^\star$. The claim is confined to this class, and the main text reports a receiver outside it, the decomposition receiver, that attains a lower error by using the shared subspace.

References

[1] Gene H. Golub, Charles F. Van Loan, *Matrix Computations*. Johns Hopkins Univ. Press, 2013.